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(54) **FRYING ROBOT**

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(57) **ABSTRACT**

A frying robot according to the present embodiment may comprise: a frame; a robot mechanism installed on the frame; and a gripper connected to the robot mechanism. The gripper may comprise: a pair of fingers for gripping an extension portion of a cooking tool; and a finger-driving mechanism which moves the pair of fingers in a direction close to or away from each other. At least one of the pair of fingers may comprise: a finger body having an insertion groove portion formed therein, in which a bent portion of the extension portion is inserted; and a fixing guide protruding from the insertion groove portion and supporting the bent portion.

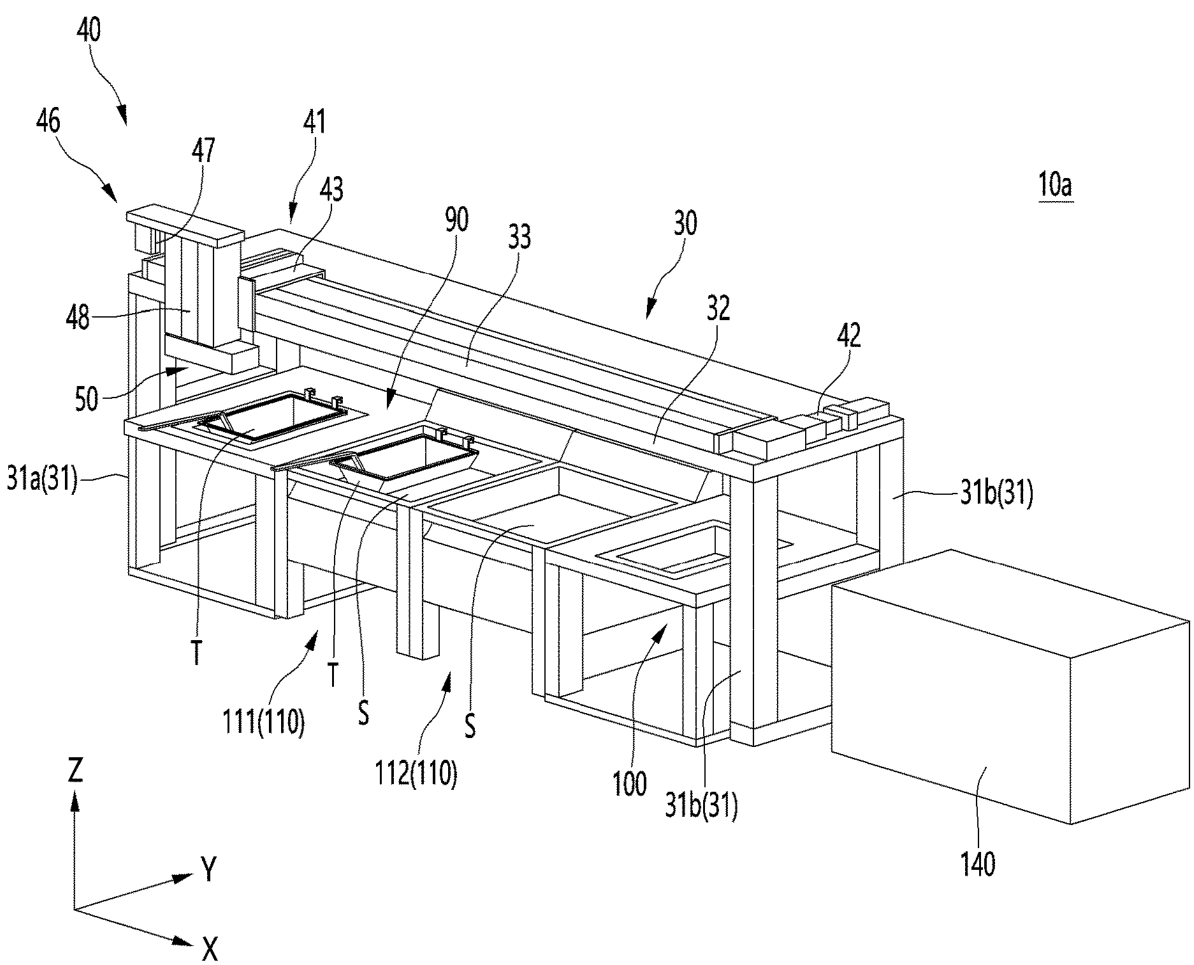


FIG. 1

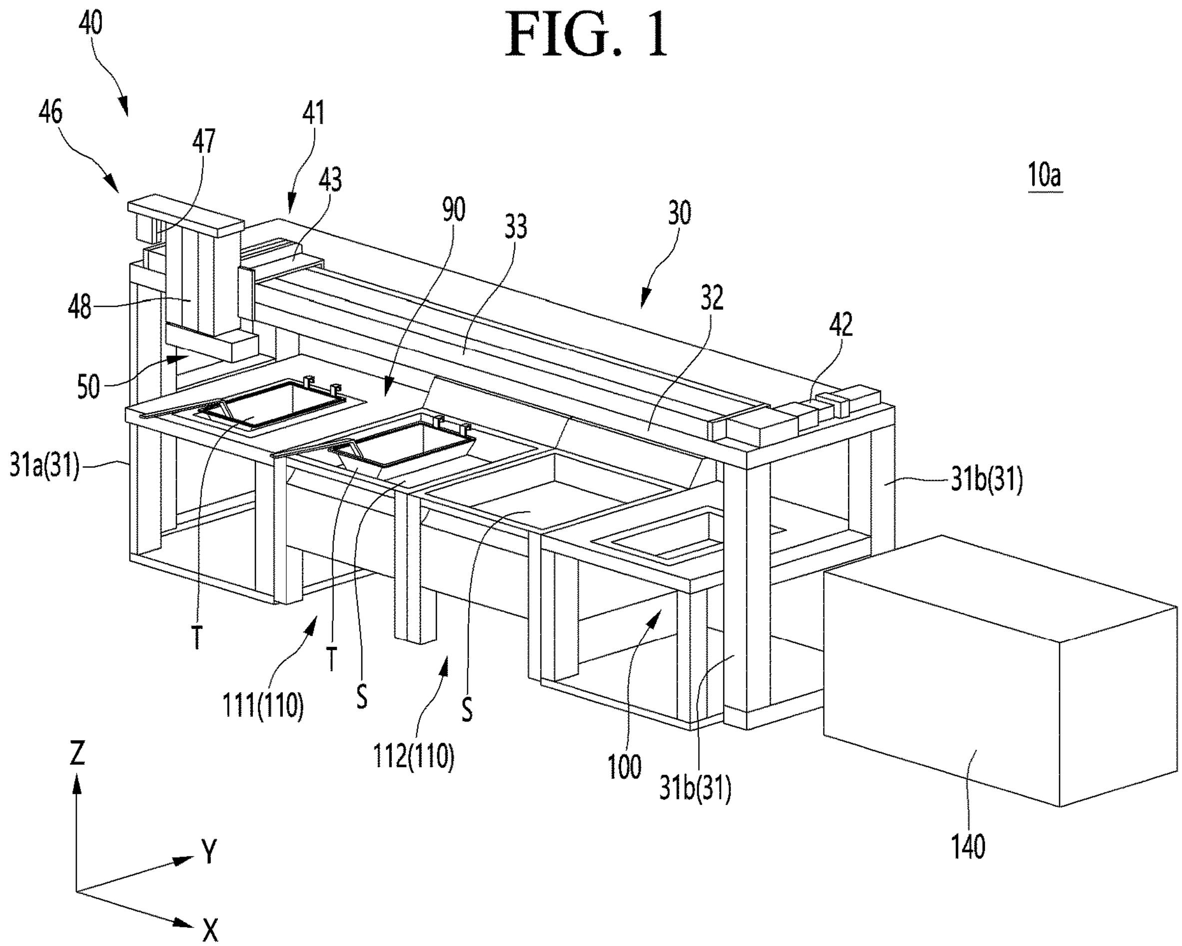

40
46
47
41
43
90
33
30
10a
48
32
42
50
31a(31)
31b(31)
T
T
111(110)
S
S
112(110)
100
31b(31)
140
Z
Y
X

FIG. 2

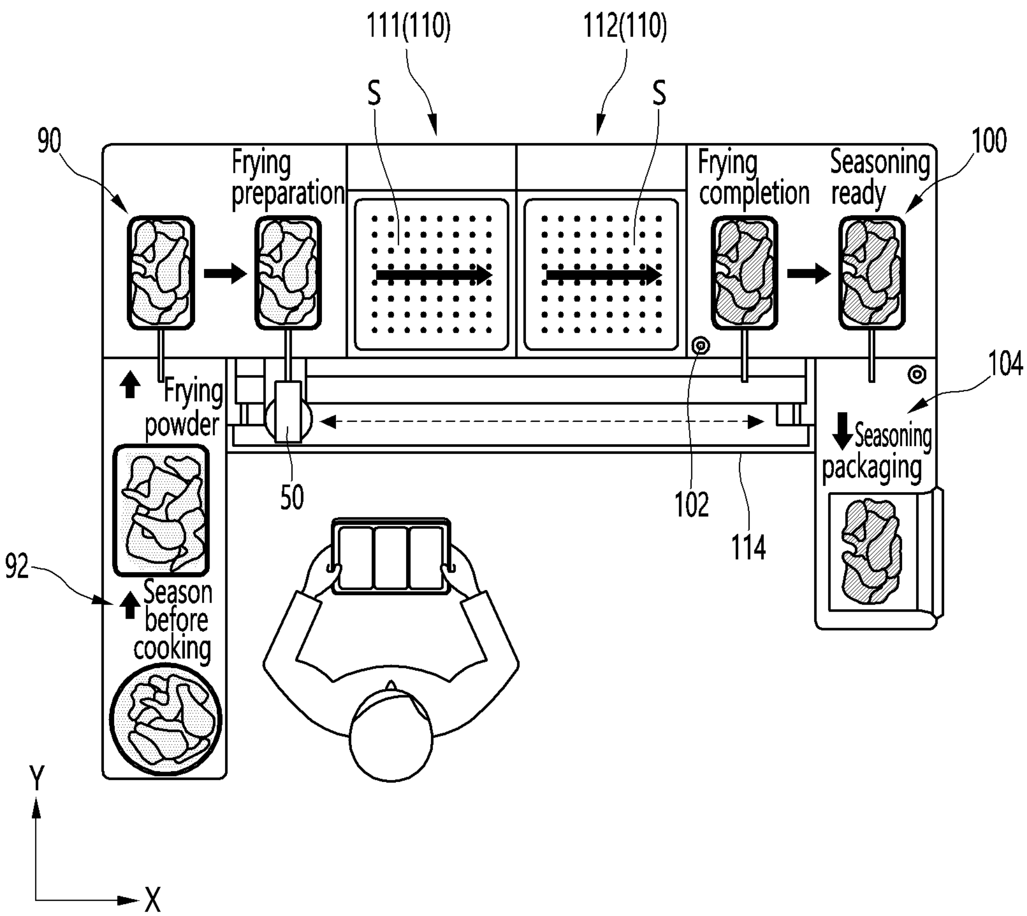

111(110)
112(110)
S
S
90
Frying preparation
Frying completion
Seasoning ready
100
104
Frying powder
Seasoning packaging
50
102
114
92
Season before cooking
Y
X

FIG. 3

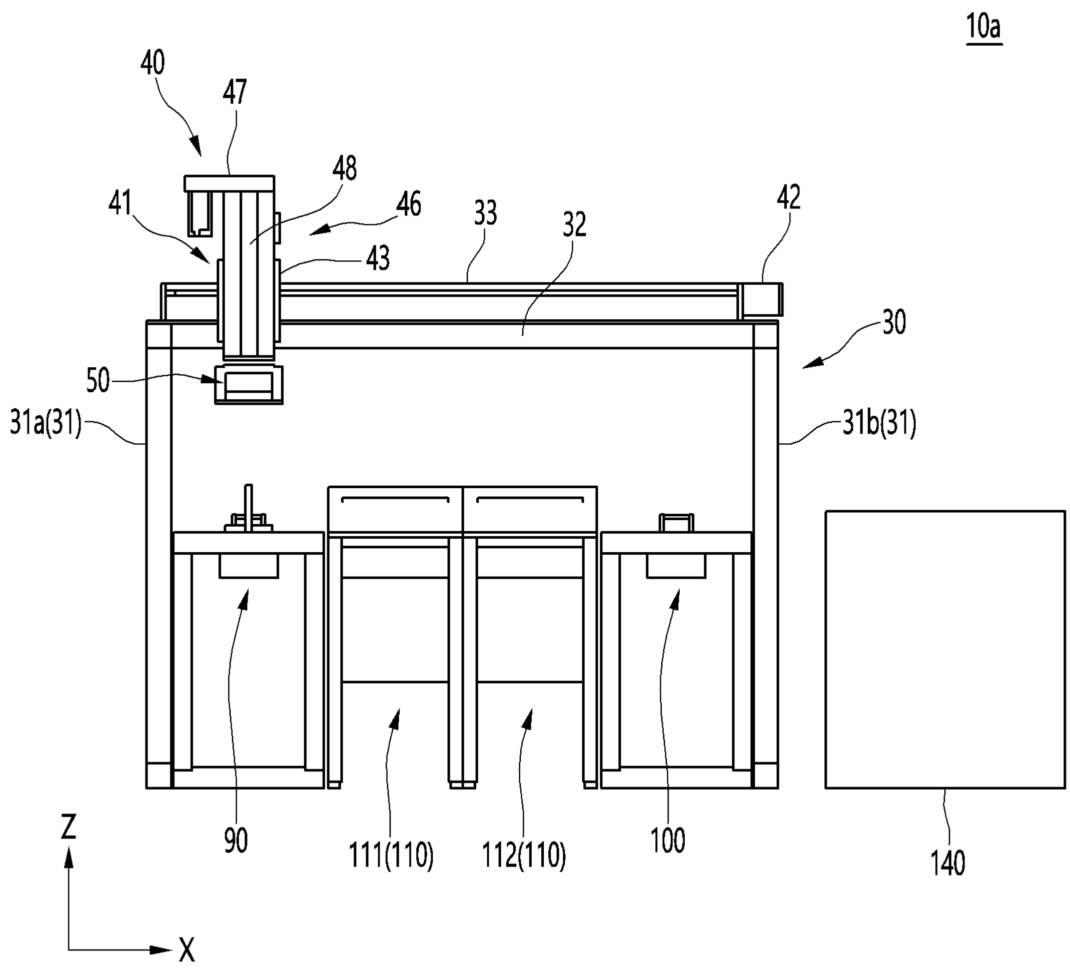

10a
40
47
48
41
46
33
32
42
43
30
50
31a(31)
31b(31)
90
111(110)
112(110)
100
140
Z
X

FIG. 4

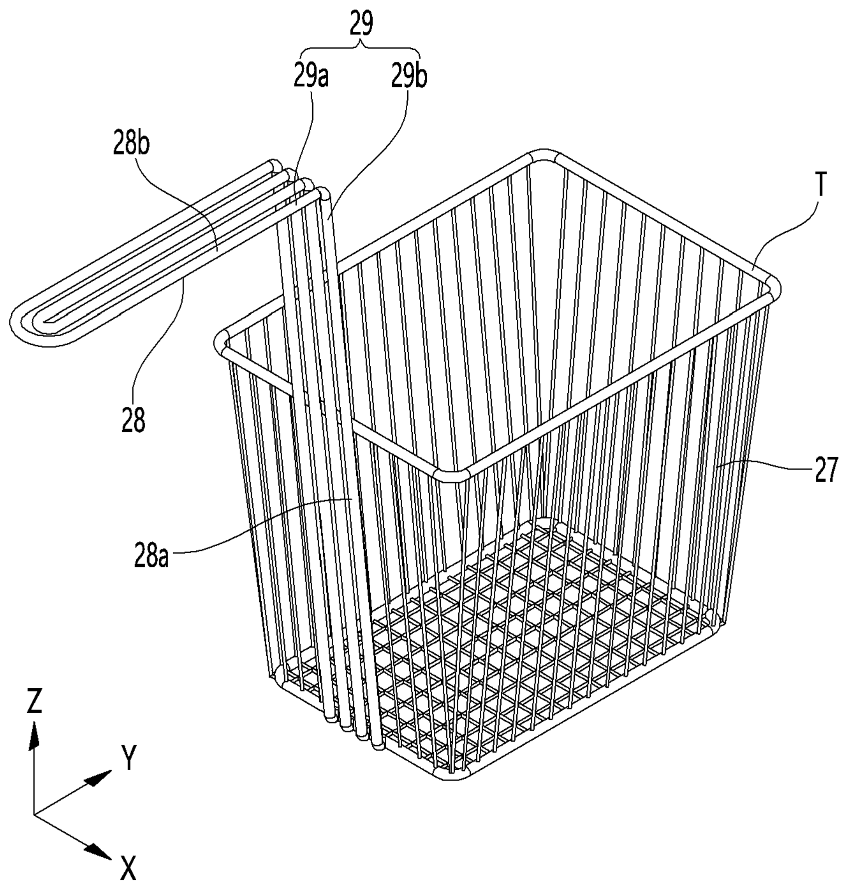


FIG. 5

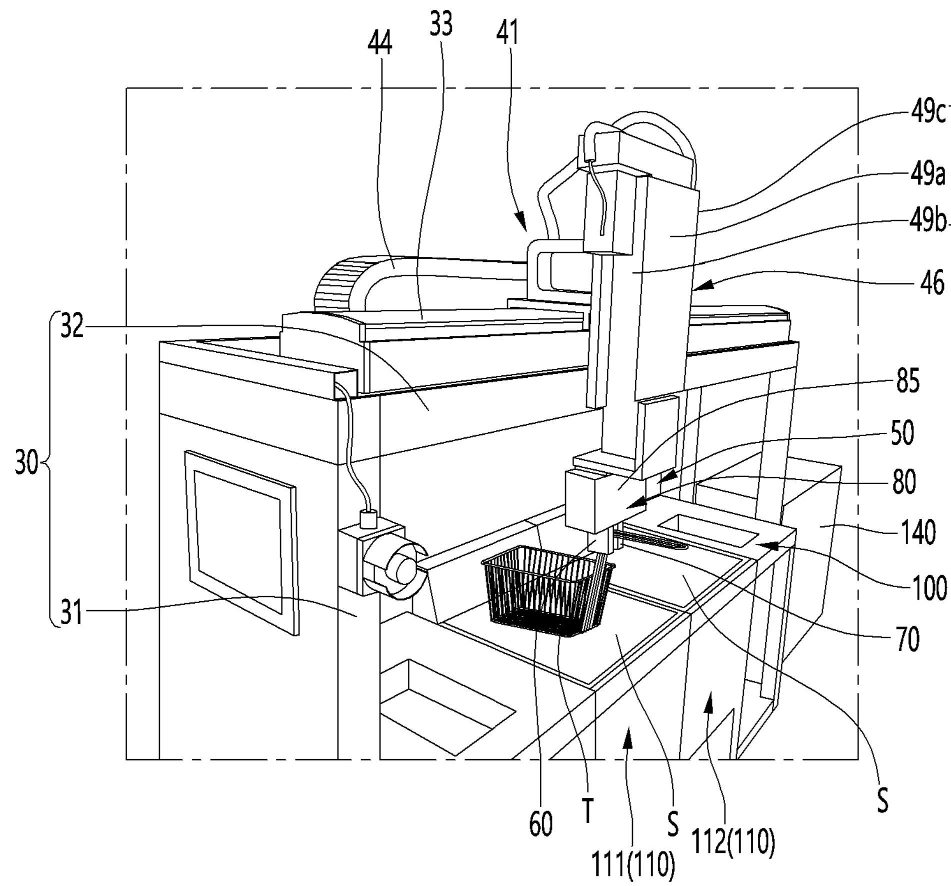

44
33
41
49c
49a
49b
49
46
32
30
31
85
50
80
140
100
70
60
T
S
111(110)
112(110)
S

FIG. 6

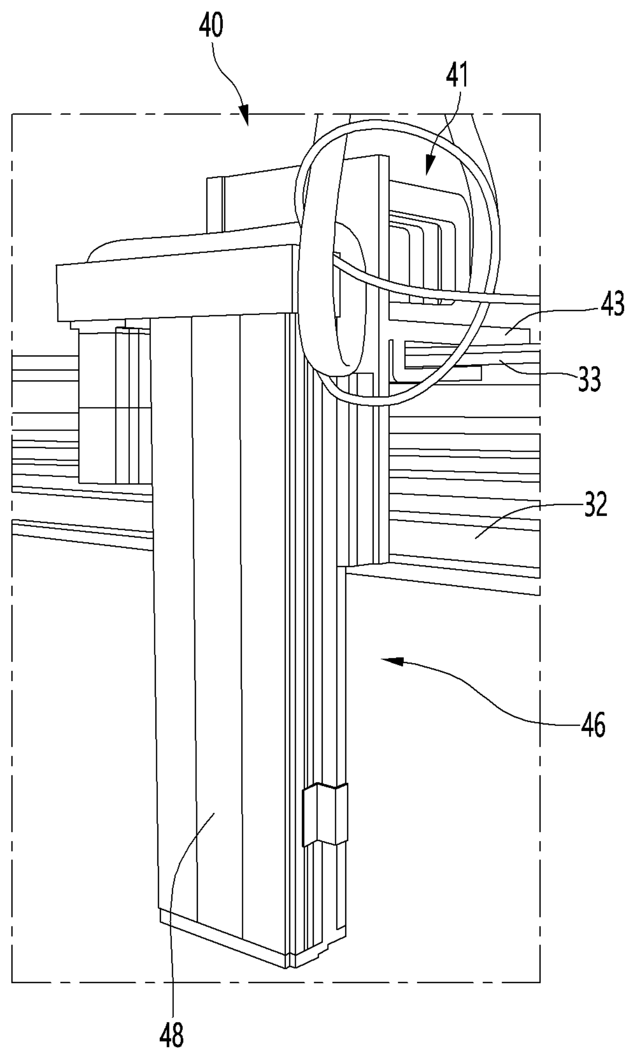


FIG. 7

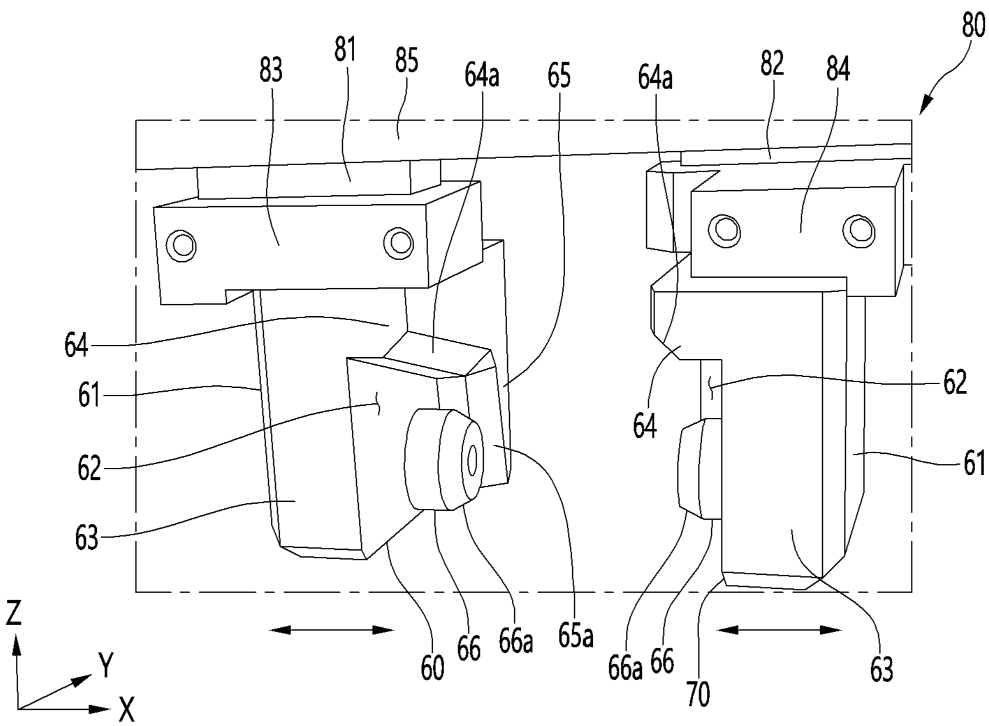

80
83
81
85
64a
65
64a
82
84
64
61
62
63
62
64
61
Z
Y
X
60
66
66a
65a
66a
66
70
63

FIG. 8

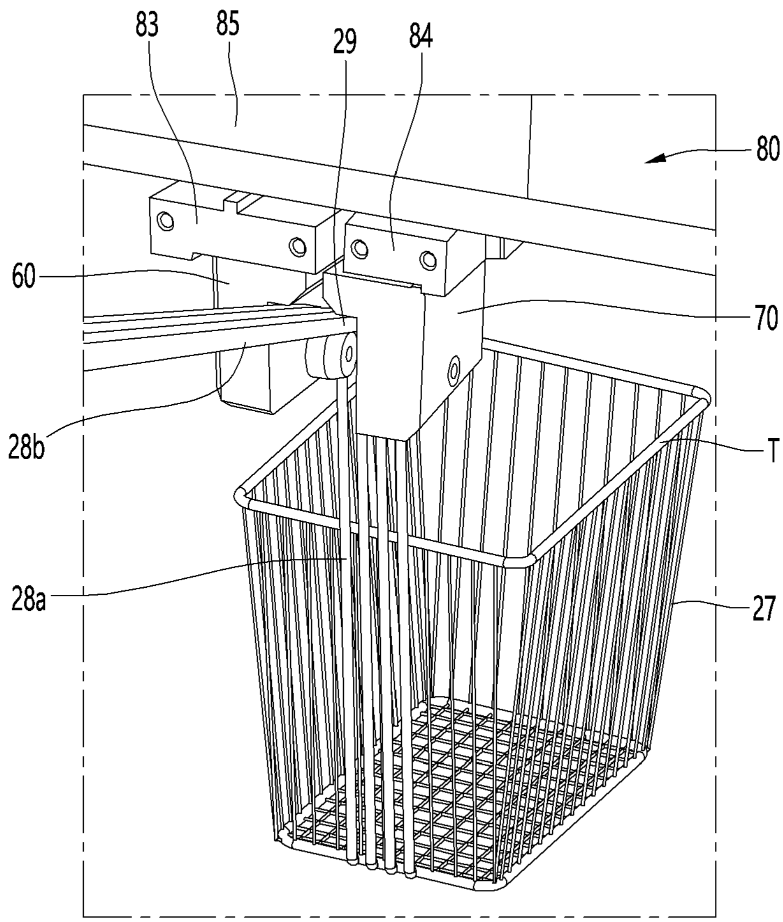


FIG. 9

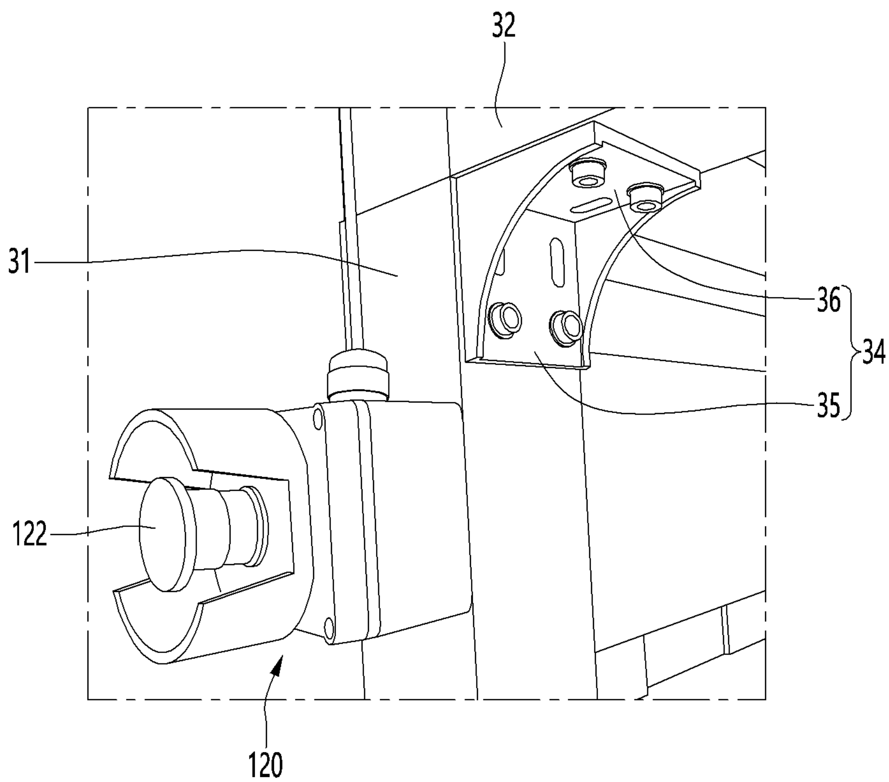


FIG. 10

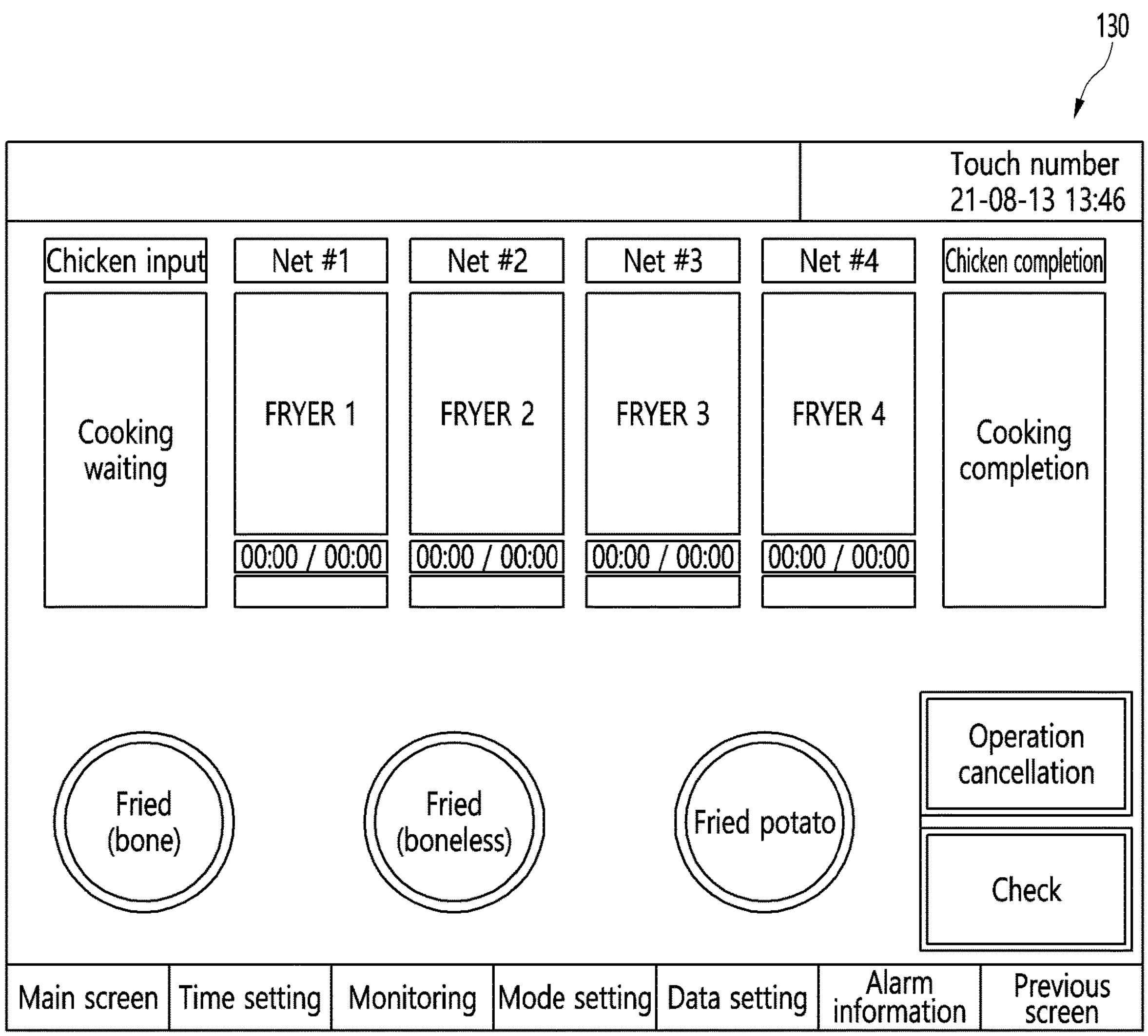

130
Touch number
21-08-13 13:46
Chicken input
Net #1
Net #2
Net #3
Net #4
Chicken completion
Cooking waiting
FRYER 1
FRYER 2
FRYER 3
FRYER 4
Cooking completion
00:00 / 00:00
00:00 / 00:00
00:00 / 00:00
00:00 / 00:00
Fried (bone)
Fried (boneless)
Fried potato
Operation cancellation
Check
Main screen
Time setting
Monitoring
Mode setting
Data setting
Alarm information
Previous screen

FIG. 11

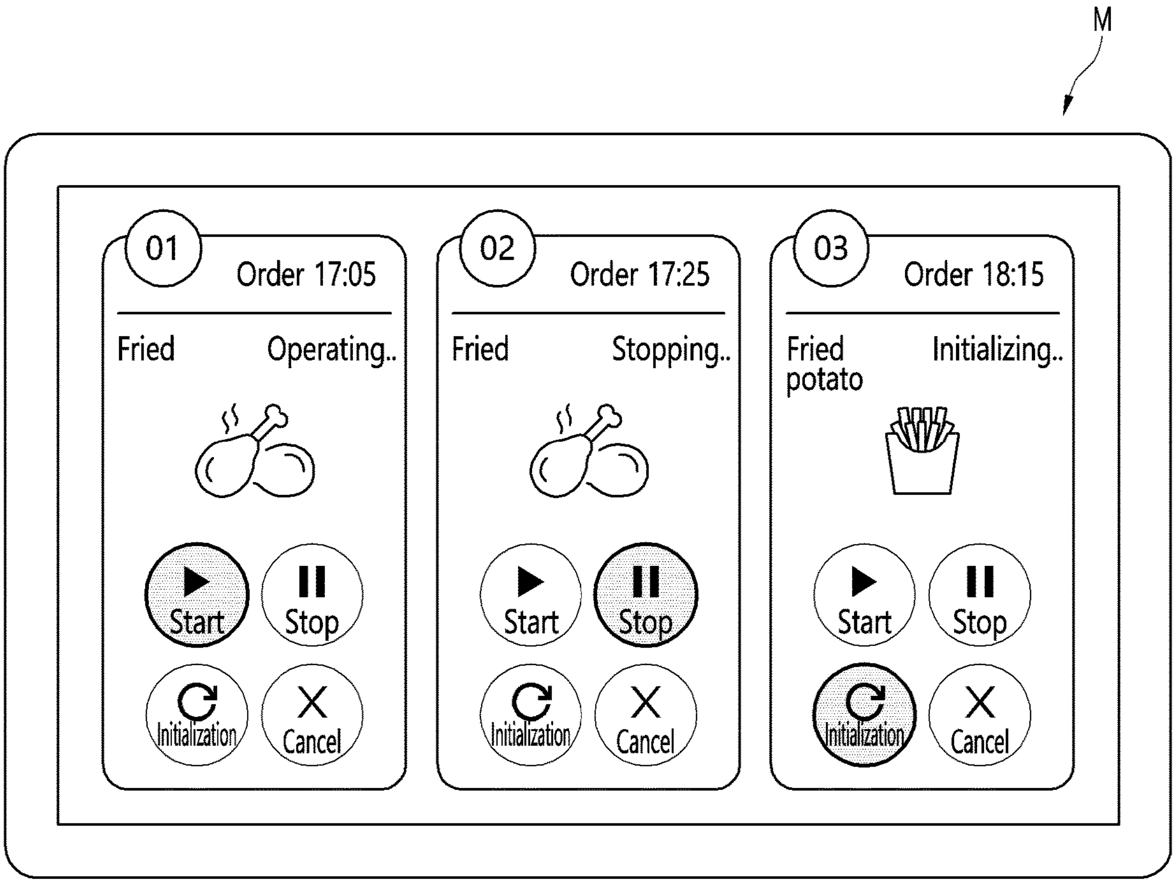

M
01
Order 17:05
Fried
Operating..
Start
Stop
Initialization
Cancel
02
Order 17:25
Fried
Stopping..
Start
Stop
Initialization
Cancel
03
Order 18:15
Fried potato
Initializing..
Start
Stop
Initialization
Cancel

FIG. 12

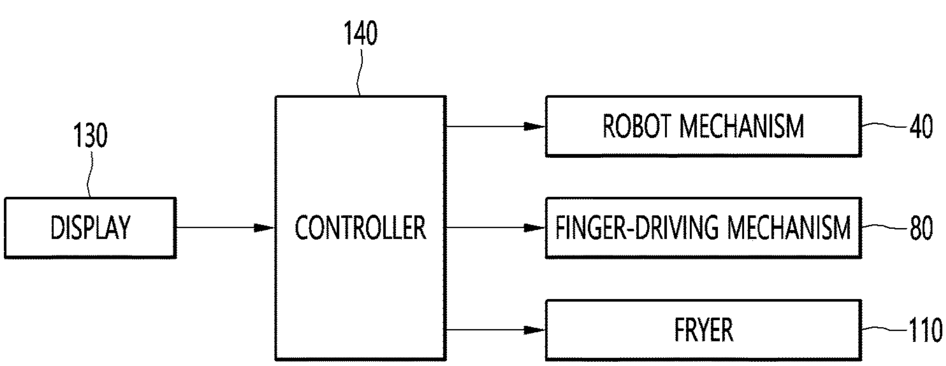
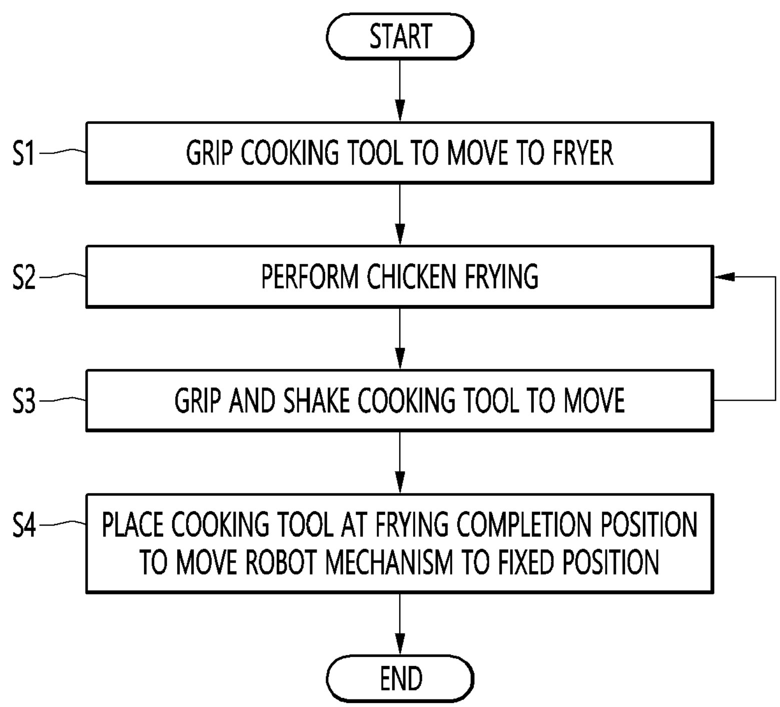

140
130
DISPLAY
CONTROLLER
ROBOT MECHANISM
40
FINGER-DRIVING MECHANISM
80
FRYER
110

FIG. 13


START
S1
GRIP COOKING TOOL TO MOVE TO FRYER
S2
PERFORM CHICKEN FRYING
S3
GRIP AND SHAKE COOKING TOOL TO MOVE
S4
PLACE COOKING TOOL AT FRYING COMPLETION POSITION TO MOVE ROBOT MECHANISM TO FIXED POSITION
END

# FRYING ROBOT

## TECHNICAL FIELD

**[0001]** The present invention relates to a robot that is capable of cooking ingredients, and more particularly, to a frying robot that is capable of frying ingredients.

## BACKGROUND ART

**[0002]** A robot is a machine that automatically processes or operates a given task based on its own abilities. The application fields of robots may be broadly classified into industrial, medical, space, underwater, etc., and may be used in a variety of fields.

**[0003]** In recent, the number of cooking robots that are capable of performing cooking using robots is gradually increasing, and an example of a robot comprises an automatic frying device disclosed in Korean Patent Publication No. 10-2004-0070450 A (published on Aug. 9, 2004). The automatic frying device comprises: a main body in which one or more sauce containers containing sauce (or flour) and at least one oil container installed to be provided with a heater for heating an oil are installed at an upper portion, and a guide groove is defined in a horizontal direction; a horizontal moving portion including a first motor installed in the main body and a horizontal movable bracket that is reciprocated in the horizontal direction by the first motor; a vertical moving part including a second motor installed on the horizontal movable bracket and a vertical movable bracket vertically elevated the second motor; and a chicken skewer portion including a vertical support that is coupled to the vertical movable bracket to pass through the guide grove so as to protrude to an upper side of the main body, an extension support that is coupled to an end of the vertical support at an upper portion of the main body to extends forward, a plurality of hooks extending downward from the extension support, and a skewer that is detachably hung on the hooks and passes through a chicken.

**[0004]** Another example of the robot is disclosed in a robot system and a method for the same, which are disclosed in Korean Patent Publication No. 10-2019-0112679 A (published on Oct. 7, 2019), and the robot system comprises a robot arm that tilts a frying net in a first direction or tilts or elevates the frying net in a second direction different from the first direction; and a controller that implements a debris removal mode in which the robot arm moves and rotates the frying net to a set debris removal path through debris inside the fryer is filtered.

## DISCLOSURE OF THE INVENTION

### Technical Problem

**[0005]** The present embodiment provides a frying robot that is capable of griping a cooking tool with high stability and reliability and cooking ingredients safely.

### Technical Solution

**[0006]** A frying robot according to the present embodiment may comprise: a frame; a robot mechanism installed on the frame; and a gripper connected to the robot mechanism.

**[0007]** The gripper may comprise: a pair of fingers for gripping an extension portion of a cooking tool; and a finger-driving mechanism which moves the pair of fingers in a direction close to or away from each other.

**[0008]** At least one of the pair of fingers may comprise: a finger body having an insertion groove portion formed therein, in which a bent portion of the extension portion is inserted; and a fixing guide protruding from the insertion groove portion and supporting the bent portion.

**[0009]** The finger body may comprise: a main body portion; an upper body portion protruding from an upper portion of the main body portion and spaced apart from the fixing guide in a vertical direction; and a side body portion protruding from the main body portion and spaced apart from the fixing guide in a left and right direction.

**[0010]** An upper chamfer configured to assist an insertion of the bent portion may be provided on the upper body portion.

**[0011]** The fixing guide may protrude from the main body portion.

**[0012]** A lower chamfer configured to assist an insertion of the bent portion may be provided on the fixing guide.

**[0013]** The finger-driving mechanism may comprise: a pair of moving shafts; a first finger bracket connected to one of the pair of moving shafts and one of the pair of fingers; and a second finger bracket connected to the other one of the pair of moving shafts and the other one of the pair of fingers.

**[0014]** The robot mechanism may comprise: a first robot moving along the frame; and a second robot disposed to be elevatable on the first robot, wherein the finger-driving mechanism may be connected to the second robot.

**[0015]** The second robot may comprise: a motor; a ball screw connected to the motor; and an outer cover configured to cover the ball screw.

**[0016]** The outer cover may comprise: a center body; a front body bent from a front end of the center body; and a rear body bent from a rear end of the center body.

**[0017]** The frame may comprise: a vertical frame; a horizontal frame disposed above the vertical frame; and a reinforcement bracket coupled to each of the vertical frame and the horizontal frame.

**[0018]** A moving guide configured to guide movement of the robot mechanism may be disposed on the horizontal frame.

**[0019]** The frying robot may further comprise: a preparation table disposed on the frame to accommodate a body of the cooking tool; a completion table disposed on the frame to be spaced apart from the first table and configured to accommodate the body of the cooking tool; and at least one fryer disposed between the preparation table and the completion table and having a space into which the body of the cooking tool is inserted.

**[0020]** The frying robot may further comprise an emergency switch configured to stop the robot mechanism.

**[0021]** The frying robot may further comprise a display configured to display a cooking time of the fryer in progress.

**[0022]** The frying robot may further comprise a controller configured to control the robot mechanism, the finger-driving mechanism, and the fryer.

**[0023]** The controller may be configured to control the robot mechanism so that the cooking tool is shaken at least once during the cooking by the fryer.

**[0024]** Another example of the frying robot comprises: a pair of fingers configured to grip an extension portion of a cooking tool; a finger-driving mechanism configured to move the pair of fingers in a direction close to or away from each other, wherein at least one of the pair of fingers comprises: a finger body having an insertion groove portion defined therein, into which a bent portion of the extension portion is inserted; and a fixing guide protruding from the insertion groove portion to support the bent portion.

**[0025]** The finger body may comprise: a main body portion; an upper body portion protruding from an upper portion of the main body portion and spaced apart from the fixing guide in a vertical direction; and a side body portion protruding from the main body portion and spaced apart from the fixing guide in a left and right direction.

**[0026]** An upper chamfer configured to assist an insertion of the bent portion may be provided on the upper body portion.

**[0027]** The fixing guide may protrude from the main body portion.

**[0028]** A lower chamfer configured to assist an insertion of the bent portion may be provided on the fixing guide.

## Advantageous Effects

**[0029]** According to the present embodiment, the bent portion of the cooking tool may be stably supported between the finger body and the fixing guide to minimize the sliding of the cooking tool and grip the cooking tool with the stability.

**[0030]** In addition, the gripper may stably support the cooking tool in the two directions to minimize the shaking of the cooking tool, quickly move the cooking tool, and safely and quickly perform the cooking.

**[0031]** In addition, the upper chamfer of the upper body portion may assist the extension portion to enter the insertion groove portion and minimize the malfunction of the gripper.

**[0032]** In addition, the lower chamfer of the fixing guide may assist the extension portion to be inserted into the insertion groove portion and minimize the malfunction of the gripper.

**[0033]** In addition, the internal configuration of the second robot may be shielded by the front body and the rear body, which are disposed on the outer cover of the second robot to improve the aesthetics and maintain and cleaning of the fryer.

**[0034]** In addition, the shaking of the frame may be minimized by the reinforcement bracket, and thus, the robot mechanism may more stably operate.

**[0035]** In addition, the user may press the emergency switch to quickly stop the robot mechanism, thereby improving the safety.

**[0036]** In addition, the user may prepare the ingredients while looking at the display to enable the faster and more convenient cooking.

**[0037]** In addition, the cooking tool may be shaken during the cooking operation using the fryer to minimize the clumping of the ingredients.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0038]** FIG. **1** is a perspective view of a frying robot according to the present embodiment.

**[0039]** FIG. **2** is a plan view of the frying robot according to the present embodiment.

**[0040]** FIG. **3** is a side view of the frying robot according to the present embodiment.

**[0041]** FIG. **4** is a perspective view of a cooking tool used in the frying robot according to the present embodiment.

**[0042]** FIG. **5** is a view when the frying robot carries the cooking tool according to the present embodiment.

**[0043]** FIG. **6** is a perspective view illustrating the inside of a second robot according to the present embodiment.

**[0044]** FIG. **7** is a perspective view before a gripper grips the cooking tool according to the present embodiment.

**[0045]** FIG. **8** is a perspective view when the gripper completely grips the cooking tool according to the present embodiment.

**[0046]** FIG. **9** is a perspective view of a frame according to the present embodiment.

**[0047]** FIG. **10** is a view of a display according to the present embodiment.

**[0048]** FIG. **11** is a view of a terminal according to the present embodiment.

**[0049]** FIG. **12** is a control block diagram according to the present embodiment.

**[0050]** FIG. **13** is a flowchart illustrating an operation of a frying robot according to the present embodiment.

## MODE FOR CARRYING OUT THE INVENTION

**[0051]** Hereinafter, detailed embodiments will be described in detail with reference to the accompanying drawings.

**[0052]** Hereinafter, when an element is described as being "coupled" or "connected" to another element, it means that the two elements are directly coupled or connected, or a third element exists between the two elements, and the two elements are coupled or connected to each other by the third element. On the other hand, the "direct coupling" or "direct connecting" of one element to the other element may be understood that the third element does not exist between the two elements.

**[0053]** A robot may refer to a machine that automatically processes or operates a given task by its own ability. In particular, a robot having a function of recognizing an environment and performing a self-determination operation may be referred to as an intelligent robot.

**[0054]** An example of a robot may be a cooking robot, particularly a frying robot. Hereinafter, the frying robot will be described with reference numeral **10***a*.

**[0055]** FIG. **1** is a perspective view of a frying robot according to the present embodiment, FIG. **2** is a plan view of the frying robot according to the present embodiment, FIG. **3** is a side view of the frying robot according to the present embodiment, and FIG. **4** is a perspective view of a cooking tool used in the frying robot according to the present embodiment.

**[0056]** The frying robot **10***a* may comprise a frame **30**, a robot mechanism **40**, and a gripper **50**. The frying robot **10***a* may cook fried food and may further comprise a preparation table **90**, a completion table **100**, and a fryer **110**.

**[0057]** The frying robot **10***a* may grip a cooking tool T disposed on the preparation table **90** to insert the cooking tool T into the fryer **110**, and feed ingredients such as raw chicken or potatoes contained in the cooking tool T may be fried in the fryer **110** in a state of being contained in the cooking tool T. When the fryer **110** completes the cooking the ingredients, the frying robot **10***a* may grip the cooking tool T again and take the cooking tool T out of the fryer **110**. The frying robot **10***a* may move the cooking tool T to the completion table **100**.

**[0058]** An example of the cooking tool T may comprise a body **27** and an extension portion **28** extending from the body **27**, as illustrated in FIG. **4**.

**[0059]** The body **27** may accommodate ingredients such as raw chicken or potatoes and may be provided in a three-dimensional shape. An example of the body **27** may be a frame. A plurality of openings may be defined in the frame, and an oil in the fryer **110** may flow into the inside of the frame through the plurality of openings, and an oil inside the frame may flow out of the frame through the plurality of openings.

**[0060]** The extension portion **28** may be provided to protrude from the body **27** and may not be inserted into the fryer **110**.

**[0061]** A handle portion may be provided on the extension portion **28**.

**[0062]** A bent portion **29** may be provided at one side of the extension portion **28**. The extension portion **28** may comprise a first extension portion **28***a*, a bent portion **29**, and a second extension portion **28***b*.

**[0063]** The first extension portion **28***a* may extend from the body **27**.

**[0064]** One end of the bent portion **29** may extend from the first extension portion **28***a*.

**[0065]** The second extension portion **28***b* may extend from the bent portion **29**.

**[0066]** An upper portion **29***a* of the bent portion **29** may be bent perpendicular to a lower portion **29***b* of the bent portion **29**, and the upper portion **29***a* of the bent portion **29** may be bent at a right angle to the lower portion **29***b* of the bent portion **29** and may also be bent to have an obtuse inclination angle.

**[0067]** The lower portion **29***b* of the bent portion **29** may extend upward from the first extension portion **28***a*.

**[0068]** The second extension portion **28***b* may extend horizontally or obliquely from the upper portion **29***a* of the bent portion **29**.

**[0069]** As illustrated in FIG. **1**, the frame **30** may be disposed to be elongated in a forward and backward direction X. The frame **30** may support the robot mechanism **40**.

**[0070]** The frame **30** may comprise a vertical frame **31** and a horizontal frame **32** disposed above the vertical frame **31**.

**[0071]** The vertical frame **31** may be provided in plurality. The vertical frame **31** may comprise a front frame **31***a* in a material movement direction and a rear frame **31***b* disposed behind the front frame **31***a*.

**[0072]** The front frame **31***a* and the rear frame **31***b* may be spaced apart from each other in the forward and backward direction X.

**[0073]** The front frame **31***a* may be provided in a pair, and the pair of front frames **31***a* may be spaced apart from each other in a left and right direction Y.

**[0074]** The rear frame **31***b* may be provided in a pair, and the pair of rear frames **31***b* may be spaced apart from each other in the left and right direction Y.

**[0075]** The horizontal frame **32** may be provided to be elongated in a longitudinal direction X of the frame **30**.

**[0076]** The horizontal frame **32** may be provided in plurality.

**[0077]** A movement guide **33** that guides movement of the robot mechanism **40** may be disposed on the horizontal frame **32**. The movement guide **33** may be disposed on an upper portion of the horizontal frame **32**. The movement guide **33** may be provided to be elongated in the longitudinal direction X of the frame **30**.

**[0078]** The robotic mechanism **40** may be disposed on the frame **30**.

**[0079]** An example of the robotic mechanism **40** may be a Cartesian type robot. The robot mechanism **40** may also be provided as a cylindrical type robot or a Scara robot. When the robot device **30** is an articulated robot mechanism such as the Scara robot, a structure of the articulated robot may be complicated and expensive, and thus, it may be preferable to be provided as a cheaper robot than the articulated robot mechanism.

**[0080]** The robotic mechanism **30** may be provided as a robot with two degrees of freedom and may move along the frame **30** in the longitudinal direction X of the frame **30** and be preferably elevated in a vertical direction Z.

**[0081]** The robotic mechanism **40** may comprise a first robot **41** that moves along the frame **30** and a second robot **46** disposed to be elevatable on the first robot **41**.

**[0082]** The first robot **41** may be a robot that moves in the longitudinal direction X of the frame **30** and may be an X-axis robot.

**[0083]** The first robot **41** may comprise a drive source **42** such as a motor (see FIGS. **1** and **3**), and a slider **43** (see FIGS. **1** and **3**) that moves along the movement guide **33**.

**[0084]** An example of the drive source **42** may comprise a motor such as linear motors or a server motor.

**[0085]** The first robot **31** may further comprise a power transmission member **44** (see FIG. **5**) connecting the drive source **42** to the slider **43**. The power transmission member **44** may comprise a linear guide connected to the drive source **42** and the slider **43**.

**[0086]** When the drive source **42** is driven, the linear guide may move the slider **43**, and the slider **43** may be guided by the movement guide **33** to move in the longitudinal direction X of the frame **30**.

**[0087]** The second robot **46** may be a robot that is elevated in the direction Z perpendicular to the longitudinal direction X of the frame **30** and may be a Z-axis robot.

**[0088]** The second robot **46** may comprise a motor **47** (see FIG. **1**), a robot body **48**, and a ball screw (not shown) connected to the motor **47**.

**[0089]** An example of the motor **47** may be a servo motor. The motor **47** may be mounted on the robot body **48**.

**[0090]** The ball screw may be connected to the motor **47** to convert rotational movement of the motor **47** into linear movement of the gripper **50**.

**[0091]** The ball screw may comprise a screw connected to the motor **47** and having a male screw disposed on an outer circumference thereof and a screw body on which a female screw engaged with the male screw is disposed on an inner circumference thereof. The screw body may be connected directly to the first robot **41** or may be connected to the first robot **41** through a separate connector.

**[0092]** The gripper **50** may be connected to the robotic mechanism **40**. The gripper **50** may be connected to the second robot **46**, may move in the longitudinal direction X of the frame **30** together with the second robot **46**, and may be elevated in the vertical direction Z together with the second robot **46**.

**[0093]** The gripper **50** may grip the bent portion **29** (see FIG. **4**) and also may grip the extension portion **28** so that sliding of the extension portion **28** (see FIG. **4**) is minimized.

**[0094]** The preparation table **90** may be disposed on frame **30**. A space may be defined in the preparation table **90** to accommodate the body **28** of the cooking tool T. The preparation table **90** may be provided as an assembly of a plurality of members.

**[0095]** An auxiliary table **92** (see FIG. **2**) may be provided around the preparation table **90**.

**[0096]** The auxiliary table **92** may be a table on which a user fries the ingredients such as the raw chicken or potatoes, and frying powder is applied to the ingredients. The auxiliary table **92** may be disposed next to the preparation table **90**.

**[0097]** The completion table **100** may be disposed on the frame **30** so as to be spaced apart from the preparation table **90**. A space may be defined in the completion table **100** to accommodate the body **28** of the cooking tool T. The completion table **100** may be provided as an assembly of a plurality of members.

**[0098]** A sensor **102** capable of sensing the cooking tool T may be disposed on the completion table **100**, and when the robot mechanism **40** moves the cooking tool T to the completion table **100**, the sensor **102** may sense the cooking tool T.

**[0099]** A packaging table **104** may be disposed around the completion table **100**.

**[0100]** The packaging table **104** may be a table on which an operator packages cooked food.

**[0101]** The fryer **110** may be disposed between the preparation table **90** and the completion table **100**. A space S may be defined in the fryer **110** into which the body **28** of the cooking tool T is inserted. At least one fryer **110** may be provided, and the plurality of fryers **110** may be disposed between the preparation table **90** and the completion table **100**.

**[0102]** The oil may be contained in the space S defined in the fryer **110**, and a heating unit of the fryer **110** may heat the oil.

**[0103]** The plurality of fryers **111** and **112** may be disposed in a row between the preparation table **90** and the completion table **100**. The preparation table **90**, the plurality of fryers **111** and **112**, and the completion table **100** may be disposed in a row in the longitudinal direction X of the frame **30**.

**[0104]** The frying robot **100*a*** may further comprise a fence **114** (see FIG. **2**) disposed to be elongated along the preparation table **90**, the at least one fryer **110**, and the completion table **100**.

**[0105]** FIG. **5** is a view when the frying robot carries the cooking tool according to the present embodiment, and FIG. **6** is a perspective view illustrating the inside of the second robot according to the present embodiment.

**[0106]** The second robot **46** may comprise an outer cover **49** (see FIG. **5**).

**[0107]** The outer cover **49** may define an outer appearance of the second robot **46** and protect internal components of the second robot **46**.

**[0108]** The outer cover **49** may be mounted on the robot body **48** to cover the ball screw, etc.

**[0109]** The outer cover **49** may comprise a center body **49*a***, a front body **49*b*** bent from a front end of the center body **49*a***, and a rear body **49*c*** bent from a rear end of the center body **49*a***.

**[0110]** Scratches that may occur on an outer surface of the robot body **48** may be minimized by the outer cover **49**.

**[0111]** As illustrated in FIG. **5**, the gripper **50** may comprise a pair of fingers **60** and **70** and a finger-driving mechanism **80**.

**[0112]** FIG. **7** is a perspective view before the gripper grips the cooking tool according to the present embodiment, and FIG. **8** is a perspective view when the gripper completely grips the cooking tool according to the present embodiment.

**[0113]** The pair of fingers **60** and **70** may grip the extension portion **28** of the cooking tool T. The pair of fingers **60** and **70** may move by the finger-driving mechanism **80**. The pair of fingers **60** and **70** may move in a direction closer to each other to grip the cooking tool T, as illustrated in FIG. **7**. The pair of fingers **60** and **70** may move away from each other and be separated from the cooking tool T, as illustrated in FIG. **6**.

**[0114]** The pair of fingers **60** and **70** may be configured symmetrically. The pair of fingers **60** and **70** may be configured to be symmetrical to each other in the forward and backward direction.

**[0115]** At least one of the pair of fingers **60** and **70** may comprise a finger body **61** and a fixing guide **66**.

**[0116]** The finger body **61** may be provided with an insertion groove portion **62** into which the bent portion **29** of the extension portion **28** is inserted.

**[0117]** The finger body **61** may comprise a main body portion **63**, an upper body portion **64**, and a side body portion **65**.

**[0118]** The upper body portion **64** may protrude from an upper portion of the main body portion **63** and may be spaced apart from the fixing guide **66** in the vertical direction Z. An upper chamfer **64*a*** that assists the insertion of the bent portion **29** may be provided on the upper body portion **64**.

**[0119]** The side body portion **65** may protrude from the main body portion **63** and may be spaced apart from the fixing guide **66** in the left and right direction Y. A side chamfer **65*a*** that assists the insertion of the bent portion **29** may be provided on the side body portion **65**.

**[0120]** The fixing guide **66** may protrude from the insertion groove portion **62** to support the bent portion **29**.

**[0121]** The fixing guide **66** may protrude from the main body portion **63**. The fixing guide **66** may be provided in a cylindrical shape. A lower chamfer **66*a*** that assists the insertion of the bent portion **29** may be provided on the fixing guide **66**.

**[0122]** The insertion groove portion **62** of any one **60** of the pair of fingers **60** and **70** and the insertion groove portion **62** of the other one **70** of the pair of fingers **60** and **70** may face each other, and the bent portion **29** of the cooking tool T may be inserted into the insertion groove **62** of each of the pair of fingers **60** and **70** when the pair of fingers **60** and **70** move closer to each other.

**[0123]** When the bent portion **29** is inserted into the insertion groove portion **62** of each of the pair of fingers **60** and **70**, an upper portion **29*a*** of the bent portion **29** may be inserted between the upper body portion **64** and the fixing guide **66**, and the upper chamfer **64*a*** and the lower chamfer **66*a*** may assist the upper portion **29*a*** of the bent portion **29** to be inserted into the insertion groove portion **62**.

**[0124]** When the bent portion **29** is inserted into the insertion groove portion **62** of each of the pair of fingers **60** and **70**, a lower portion **29*a*** of the bent portion **29** may be inserted between the side body portion **64** and the fixing guide **66**, and the side chamfer **65*a*** and the lower chamfer

**66***a* may assist the lower portion **29***a* of the bent portion **29** to be inserted into the insertion groove portion **62**.

**[0125]** The finger-driving mechanism **80** may move the pair of fingers **60** and **70** in a direction closer to each other, as illustrated in FIG. **7**, and may move the pair of fingers **60** and **70** in a direction away from each other, as illustrated in FIG. **6**.

**[0126]** The finger-driving mechanism **80** may be connected to the second robot **46**.

**[0127]** As illustrated in FIG. **6**, the finger-driving mechanism **80** may comprise a pair of moving shafts **81** and **82**, a first finger bracket **83**, and a second finger bracket **84**.

**[0128]** The pair of moving shafts **81** and **82** may be spaced apart from each other in the longitudinal direction X of the frame.

**[0129]** The first finger bracket **83** may be connected to one **81** of the pair of moving shafts **81** and **82** and one **60** of the pair of fingers **60** and **70**.

**[0130]** One **60** of the pair of fingers **60** and **70** may be coupled to the first finger bracket **83** through a coupling member such as a screw.

**[0131]** The second finger bracket **84** may be connected to the other one **82** of the pair of moving shafts **81** and **82** and the other one **70** of the pair of fingers **60** and **70**.

**[0132]** The other one **70** of the pair of fingers **60** and **70** may be coupled to the second finger bracket **84** through a coupling member such as a screw.

**[0133]** The finger-driving mechanism **80** may further comprise a driver housing **85** connected to a lower portion of the second robot **46**.

**[0134]** A pair of moving shafts **81** and **82** may be movably disposed on a lower portion of the driver housing **85**.

**[0135]** A first advance/retract mechanism (not shown) that moves one **81** of the pair of moving shafts **81** and **82**, and a second advance/retract mechanism (not shown) that moves the other one **82** of the pair of moving shafts **81** and **82** may be disposed inside the driver housing **85**.

**[0136]** The first advance/retract mechanism may comprise a first drive source such as a motor, and a power transmission member connecting any one **81** of the pair of moving shafts **81** and **82** to the first drive source.

**[0137]** The second advance/retract mechanism may comprise a second drive source such as a motor, and a power transmission member connecting the other one **82** of the pair of moving shafts **81** and **82** to the second drive source.

**[0138]** FIG. **9** is a perspective view of a frame according to the present embodiment.

**[0139]** The frame **30** may comprise reinforcement brackets **34** that are respectively coupled to a vertical frame **31** and a horizontal frame **32**.

**[0140]** The reinforcement bracket **34** may comprise a vertical portion **35** coupled to the vertical frame **31** through a coupling member such as a screw, and a vertical portion **36** coupled to the horizontal frame **32** through a coupling member such as a screw.

**[0141]** The reinforcement brackets **34** may be provided in plurality.

**[0142]** The frying robot may further comprise an emergency switch **120** that stops the robot mechanism **40**. The emergency switch **120** may comprise a button **122** pressed or turned by the user.

**[0143]** When the user presses or turns the button **122** of the emergency switch **120**, power supplied from a power supply of the frying robot **10***a* to the robot mechanism **40** may be cut off, and the user may respond to an emergency situation.

**[0144]** FIG. **10** is a view of a display according to the present embodiment.

**[0145]** The frying robot **10***a* may further comprise a display **130** that displays a cooking time of the fryer **110** in progress.

**[0146]** The display **130** may be disposed on a front or side surface of the frame **30**.

**[0147]** The display **130** may display fryers **110** in progress of cooking and may display a remaining cooking time of each fryer **110**.

**[0148]** The display **130** may display a bone-in fried food selection menu, a boneless fried food selection menu, a menu of deep-fried food, etc.

**[0149]** The user may input the fryer to be cooked, selection of the cooking menu, operation cancellation, lock, etc. through the display **130**, and may check the fryer in progress of cooking, the selected cooking menu, the operation cancellation, the lock, etc. through the display **130**.

**[0150]** FIG. **11** is a view of a terminal according to the present embodiment, FIG. **12** is a control block diagram according to the present embodiment, and FIG. **13** is a flowchart illustrating an operation of a frying robot according to the present embodiment.

**[0151]** The frying robot **10***a* may further comprise a terminal M such as a tablet. The terminal M may be connected to a controller **140** through a communication means such as Wi-Fi.

**[0152]** The terminal M such as a tablet may provide an order sequence and an order time, display menu names, operation status, and menu images, and provide operation manipulation buttons for start/stop/reset/delete.

**[0153]** The user may check the order sequence, the order time, the menu names, the operation status, and menu for inputting an operation command of the frying robot **10***a* through the operation manipulation button.

**[0154]** The frying robot may further comprise a controller **140**. The controller **140** may control the robot mechanism **40**, the finger-driving mechanism **80**, and the fryer **110**.

**[0155]** The controller **140** may control the robot mechanism **40** so that the cooking tool Tis shaken at least once during the cooking by the fryer **110**.

**[0156]** The frying robot **10***a* may grip the cooking tool T disposed on the preparation table **90** and then insert a net of the cooking tool T into the space S of the fryer **110**. (S**1**)

**[0157]** After the net **27** of the cooking tool T is inserted into the space S**2** of the fryer **110**, the fryer **110** may perform cooking (frying). (S**2**)

**[0158]** When a set time has elapsed while the fryer **110** is frying ingredients contained in the net **27**, the frying robot **10***a* may grip a bent portion **29** of the cooking tool T to shake the net **27** at least once. (S**3**)

**[0159]** The frying robot **10***a* may shake the net **27** at least once, and when a completion set time is reached, the frying robot **10***a* may control a robot mechanism **40** so that the net **27** is disposed on a completion table **100**, control the robot mechanism **40** and a finger-driving mechanism **80** so that the net **27** is disposed on the completion table **100**, and move the robot mechanism **40** to a fixed position. (S**4**)

**[0160]** The above-disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments, which fall within the true spirit and scope of the present invention.

[0161] Thus, the embodiment of the present invention is to be considered illustrative, and not restrictive, and the technical spirit of the present invention is not limited to the foregoing embodiment.

[0162] Therefore, the scope of the present invention is defined not by the detailed description of the invention but by the appended claims, and all differences within the scope will be construed as being comprised in the present invention.

**1**. A frying robot comprising:
a frame;
a robot mechanism installed on the frame; and
a gripper connected to the robot mechanism,
wherein the gripper comprises:
a pair of fingers configured to grip an extension portion of a cooking tool;
a finger-driving mechanism configured to move the pair of fingers in a direction close to or away from each other,
wherein at least one of the pair of fingers comprises:
a finger body having an insertion groove portion defined therein, into which a bent portion of the extension portion is inserted; and
a fixing guide protruding from the insertion groove portion to support the bent portion.

**2**. The frying robot according to claim **1**, wherein the finger body comprises:
a main body portion;
an upper body portion protruding from an upper portion of the main body portion and spaced apart from the fixing guide in a vertical direction; and
a side body portion protruding from the main body portion and spaced apart from the fixing guide in a left and right direction.

**3**. The frying robot according to claim **1**, wherein an upper chamfer configured to assist an insertion of the bent portion is provided on the upper body portion.

**4**. The frying robot according to claim **3**, wherein the fixing guide protrudes from the main body portion.

**5**. The frying robot according to claim **1**, wherein a lower chamfer configured to assist an insertion of the bent portion is provided on the fixing guide.

**6**. The frying robot according to claim **1**, wherein the finger-driving mechanism comprises:
a pair of moving shafts;
a first finger bracket connected to one of the pair of moving shafts and one of the pair of fingers; and
a second finger bracket connected to the other one of the pair of moving shafts and the other one of the pair of fingers.

**7**. The frying robot according to claim **1**, wherein the robot mechanism comprises:
a first robot moving along the frame; and
a second robot disposed to be elevatable on the first robot,
wherein the finger-driving mechanism is connected to the second robot.

**8**. The frying robot according to claim **1**, wherein the second robot comprises:
a motor;
a ball screw connected to the motor; and
an outer cover configured to cover the ball screw.

**9**. The frying robot according to claim **1**, wherein the outer cover comprises:
a center body;
a front body bent from a front end of the center body; and
a rear body bent from a rear end of the center body.

**10**. The frying robot according to claim **1**, wherein the frame comprises:
a vertical frame;
a horizontal frame disposed above the vertical frame; and
a reinforcement bracket coupled to each of the vertical frame and the horizontal frame.

**11**. The frying robot according to claim **1**, wherein a moving guide configured to guide movement of the robot mechanism is disposed on the horizontal frame.

**12**. The frying robot according to claim **1**, further comprising:
a preparation table disposed on the frame to accommodate a body of the cooking tool;
a completion table disposed on the frame to be spaced apart from the first table and configured to accommodate the body of the cooking tool; and
at least one fryer disposed between the preparation table and the completion table and having a space into which the body of the cooking tool is inserted.

**13**. The frying robot according to claim **12**, further comprising an emergency switch configured to stop the robot mechanism.

**14**. The frying robot according to claim **12**, further comprising a display configured to display a cooking time of the fryer in progress.

**15**. The frying robot according to claim **12**, further comprising a controller configured to control the robot mechanism, the finger-driving mechanism, and the fryer,
wherein the controller is configured to control the robot mechanism so that the cooking tool is shaken at least once during the cooking by the fryer.

**16**. A frying robot comprising:
a pair of fingers configured to grip an extension portion of a cooking tool;
a finger-driving mechanism configured to move the pair of fingers in a direction close to or away from each other,
wherein at least one of the pair of fingers comprises:
a finger body having an insertion groove portion defined therein, into which a bent portion of the extension portion is inserted; and
a fixing guide protruding from the insertion groove portion to support the bent portion.

**17**. The frying robot according to claim **16**, wherein the finger body comprises:
a main body portion;
an upper body portion protruding from an upper portion of the main body portion and spaced apart from the fixing guide in a vertical direction; and
a side body portion protruding from the main body portion and spaced apart from the fixing guide in a left and right direction.

**18**. The frying robot according to claim **1**, wherein an upper chamfer configured to assist an insertion of the bent portion is provided on the upper body portion.

**19**. The frying robot according to claim **3**, wherein the fixing guide protrudes from the main body portion.

**20**. The frying robot according to claim **1**, wherein a lower chamfer configured to assist an insertion of the bent portion is provided on the fixing guide.

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